Ketamine

Mechanism of action

- Ketamine has complex actions but is predominantly an N-methyl-d-aspartate (NMDA) receptor antagonist (blocker), resulting in inhibition of excitatory neurotransmitters in the brain.
- Low doses cause analgesia, larger doses cause amnesia and dissociation, and high doses cause anaesthesia.

Authorised practice levels

- Paramedics: analgesia.
- ICPs and CCPs: all indications.

Indications

- ✓ Severe pain that has not been adequately controlled with an opiate.
- ✓ Inducing dissociation.
- ✓ Acute behavioural disturbance causing a severe to immediately life-threatening risk to safety.
- ✓ Rapid sequence intubation (RSI).
- ✓ Significant movement during CPR that is interfering with resuscitation.
- ✓ Asthma with severe agitation that is impairing the ability to safely provide treatment or transport.

Contraindications

- ✗ Known severe allergy.

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Cautions

- Altered level of consciousness. Ketamine may reduce the level of consciousness.
- Signs of shock. Ketamine may make shock worse.
- Concurrent administration of opiates or midazolam. This will increase and prolong the effects.
- Aged greater than or equal to 75 years, particularly if frail. The effects of ketamine will be increased and prolonged in this cohort.
- Aged less than five years.

Use in pregnancy or when breastfeeding

- Safety has not been demonstrated during pregnancy, but ketamine should be administered if indicated.
- May be administered if the patient is breastfeeding. Advise the patient to stop breastfeeding and seek further advice from their lead maternity carer or GP.

Dosage

- For analgesia:
  - IV:
    - 0.2 mg/kg IV up to a maximum of 20 mg for an adult, administered over approximately 15 minutes. Repeat as required.
    - Consider commencing an infusion using service guidelines.
    - See the paediatric drug dose tables for a child and administer IV over approximately 15 minutes. Repeat as required.
  - IM or PO:
    - 0.5 mg/kg IM or PO for an adult, up to a maximum of 50 mg, if IV access cannot be obtained. This may be repeated once after 20 minutes.
    - See the paediatric drug dose tables for a child. This may be repeated once after 20 minutes.
- For dissociation:
  - 1 mg/kg IV (up to a maximum of 150 mg) for an adult. This may be repeated once after five minutes.
  - 5 mg/kg IM (up to a maximum of 400 mg) for an adult, once only.
  - See the paediatric drug dose tables for a child.
  - Halve all doses if a caution is present.

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- For acute behavioural disturbance:
  - 1 mg/kg IV (up to a maximum of 150 mg) every five minutes as required, or
  - 5 mg/kg IM, rounded to the nearest 100 mg (up to a maximum of 400 mg).
  - The IM dose may be repeated once after 20 minutes.
- For asthma with severe agitation: 0.5 mg/kg IV, up to a maximum of 50 mg. This may be repeated every 10 minutes as required.
- For RSI: see the **'RSI'** guideline.
- For significant movement during CPR that is interfering with resuscitation administer 1 mg/kg (up to a maximum of 150 mg) of ketamine. This may be repeated once if required.

Administration

- The preferred route of administration is IV.
- Administer analgesia doses IV over approximately 15 minutes as this appears to reduce adverse side effects.
- For analgesia: Dilute 100 mg (1 ml) of ketamine in a 100 ml bag of 5% glucose. Shake well and label:
  - For adults: draw up the dose and administer 2 mg (2 ml) every 1-2 minutes.
  - For children: draw up the dose, dilute the volume further to a total of 20 ml using 0.9% sodium chloride and administer 2 ml every 1-2 minutes.
- Administer dissociation doses by diluting 100 mg (1 ml) to a total of 10 ml using 0.9% sodium chloride and administering the dose as an IV bolus.
- The IM route is preferred over the PO route as IM absorption is more reliable. Administer IM ketamine undiluted. The preferred site is the lateral thigh as this has the best absorption. If this site is not suitable use the lateral upper arm.
- The PO route should be reserved for the very unusual circumstance in which IM injection is contraindicated or not feasible. Administer ketamine PO undiluted in a liquid, for example paracetamol syrup or water.

Common adverse effects

- Transient hypertension.
- Tachycardia.

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- Apnoea.
- Nausea and vomiting.
- Sedation.
- Hallucinations.

Usual onset of effect

- IV: 1-2 minutes.
- IM: 5-10 minutes.
- PO: 10-20 minutes.

Usual duration of effect

- 10-60 minutes.

Usual preparation

- Ampoule containing 200 mg in 2 ml.

Pharmacokinetics

- Ketamine is predominantly metabolised in the liver.
- There are no significant effects from liver impairment on acute administration.

Common interactions

- The effects will be increased in the presence of other analgesic medicines or sedatives, for example opiates, benzodiazepines or alcohol.

Additional information

- Warn the patient they may feel strange following ketamine administration.

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- Prior to administering ketamine for pain, sufficient opiates should be administered so that further doses are not providing additional analgesia. This will usually require 150-200 mcg of fentanyl or 15-20 mg of morphine for an adult.
- Do not treat hallucinations routinely with midazolam because the combination of midazolam and ketamine is commonly associated with a reduced level of consciousness, particularly if an opiate has also been administered. Most hallucinations will settle with a combination of explanation and time. However, midazolam in low doses may be administered IV if the hallucinations are severe provided the patient is obeying commands and physiologically stable.
- Round the patient's weight up to the nearest 10 kg when calculating doses.

More Information

Version history

16 June 2026 1.1.0.1
Update: Contraindications, cautions and dosage updated.

11 September 2023 1.0.4
Update: Multiple changes.

22 November 2022 1.0.1
Update: Indications, cautions, infusion instructions and aligned with analgesia CPG.